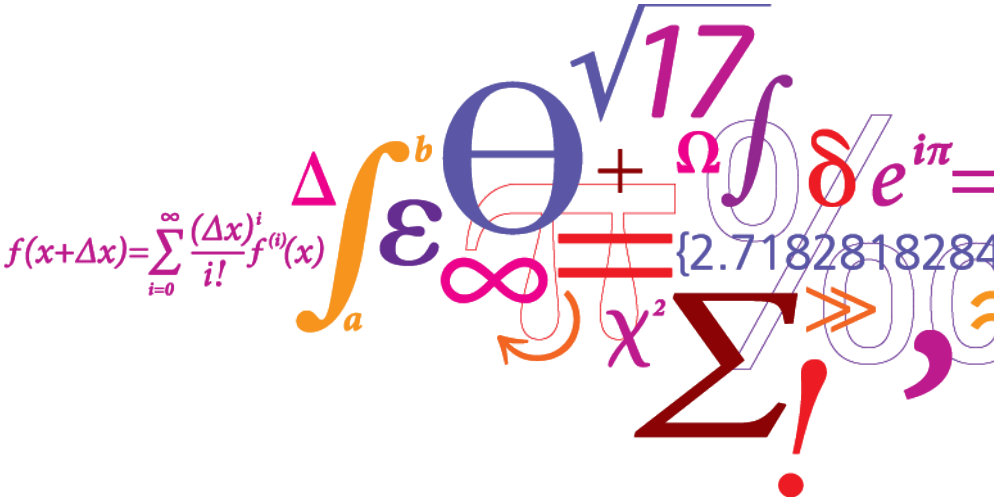
46755 – Renewables in Electricity Markets

Lecture 1: Introduction to electricity markets

Jalal Kazempour

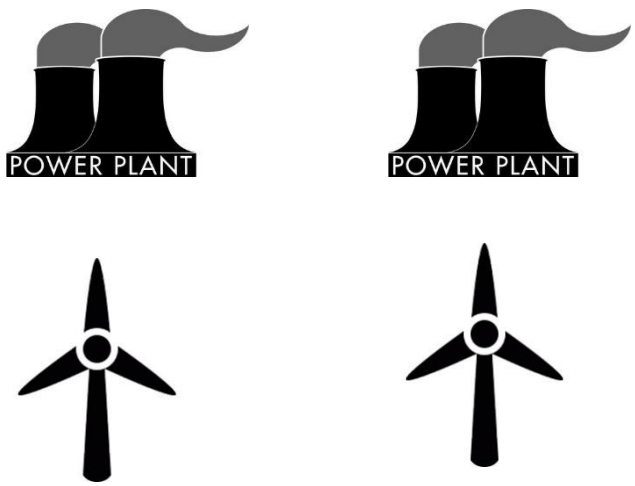
February 3, 2025



Electric Power Systems

Generators

conventional and renewable generators



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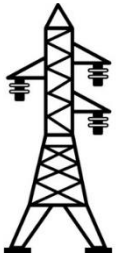
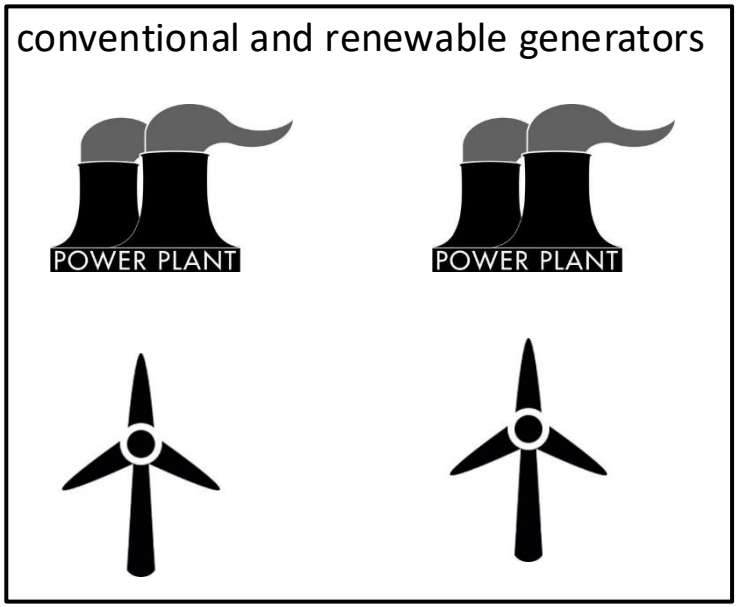


Demands

Electric demands

Electric Power Systems

Generators



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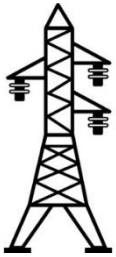


Demands

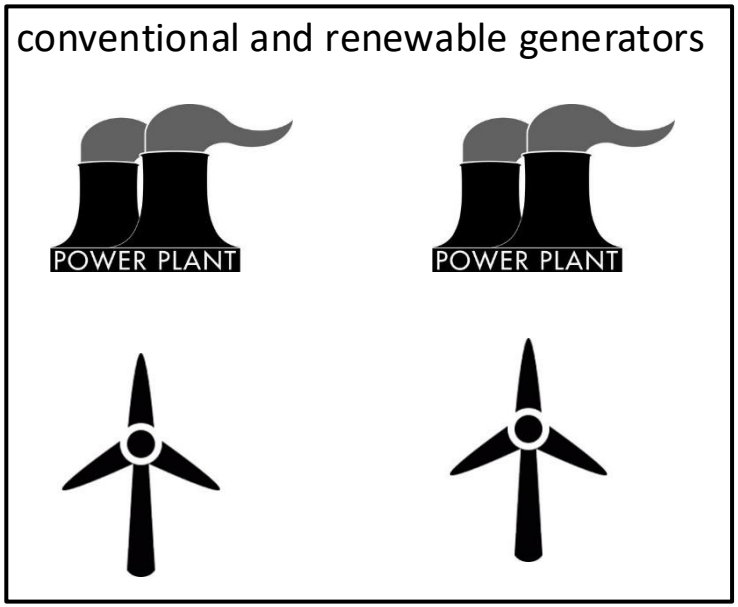
Electric demands

Organizational Structure of Electric Power Systems:

- Centralized Power Systems (no market!)
- Electricity Markets

Electric Power Systems

Generators



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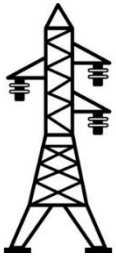


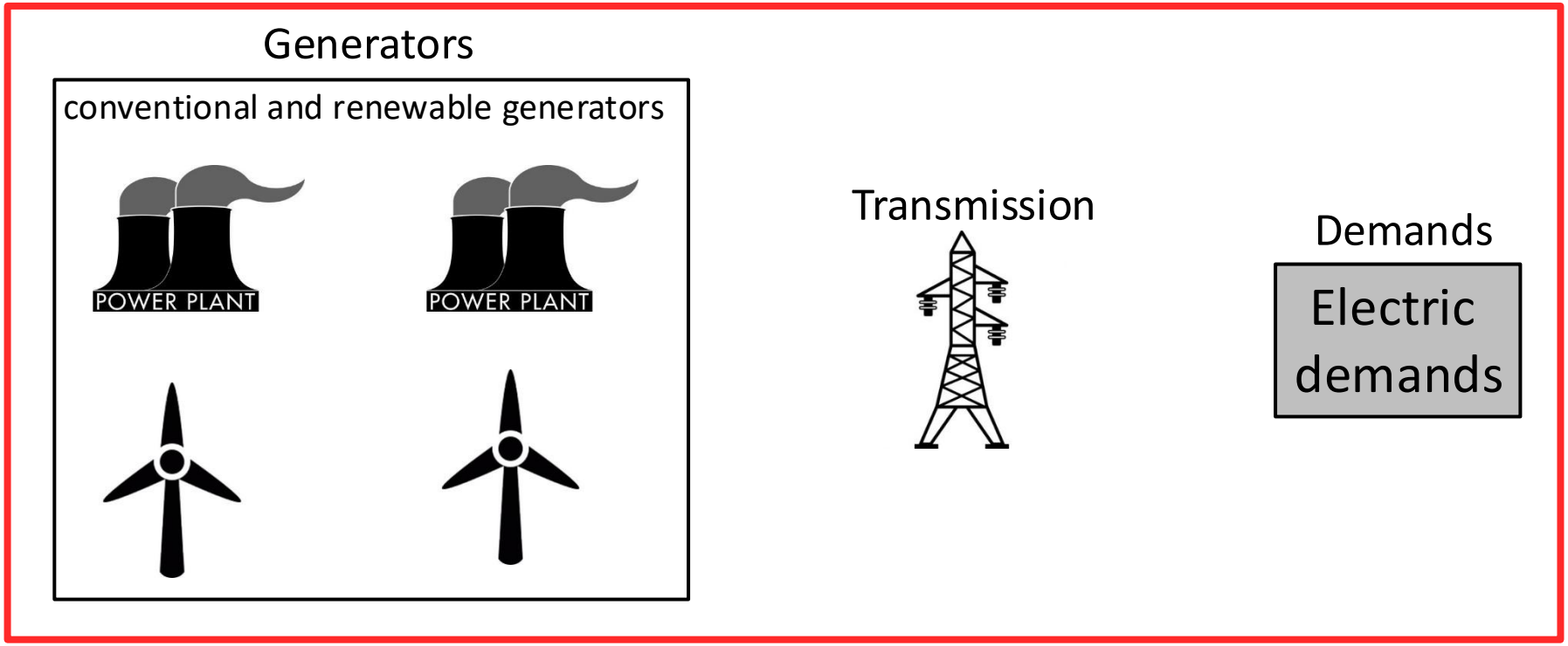
Demands

Electric demands

Organizational Structure of Electric Power Systems:

- **Centralized Power Systems (no market!)**
- Electricity Markets

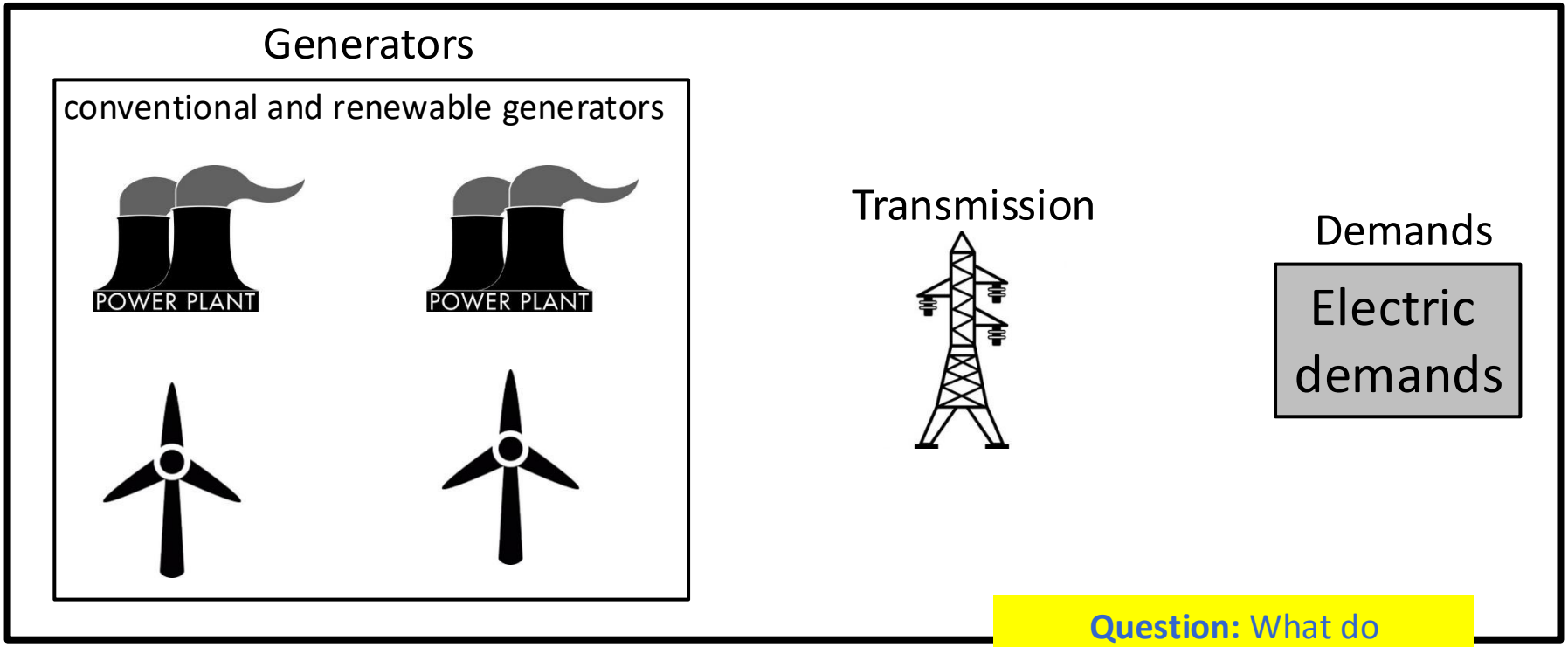
Centralized Power Systems



Centralized power system (no market):

A **<u>single</u>** entity, such as the system operator, is responsible for making all operational and planning decisions for the entire power system.

Centralized Power Systems



Question: What do "operational" and "planning" decisions mean?

Centralized power system (no market):

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Centralized Power Systems

- What is the goal of a system operator when making operational and planning decisions?

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- ✓ To meet the entire demand by dispatching power generators across the network in a **feasible** and **cost-effective** manner.

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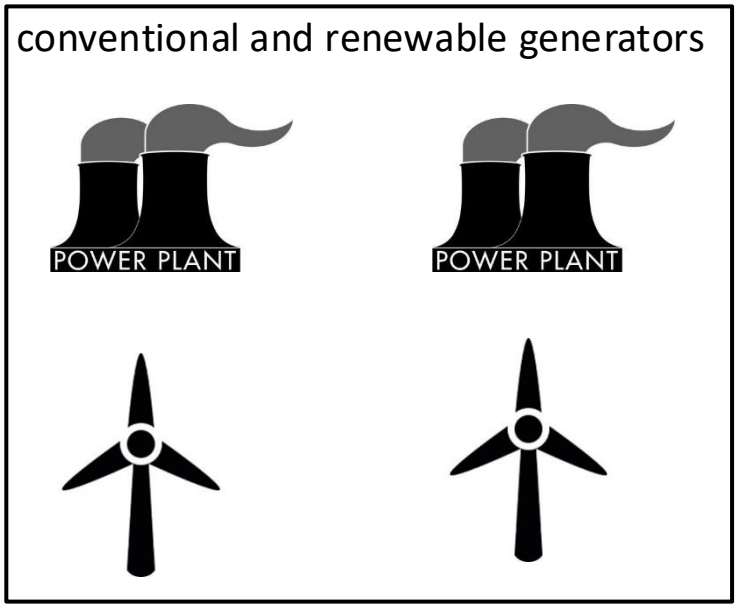
Question: What do “feasibility” and “cost-effectiveness” mean?

Electric Power Systems

Generators



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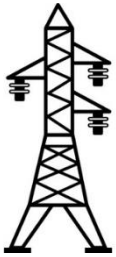

Demands

Electric demands

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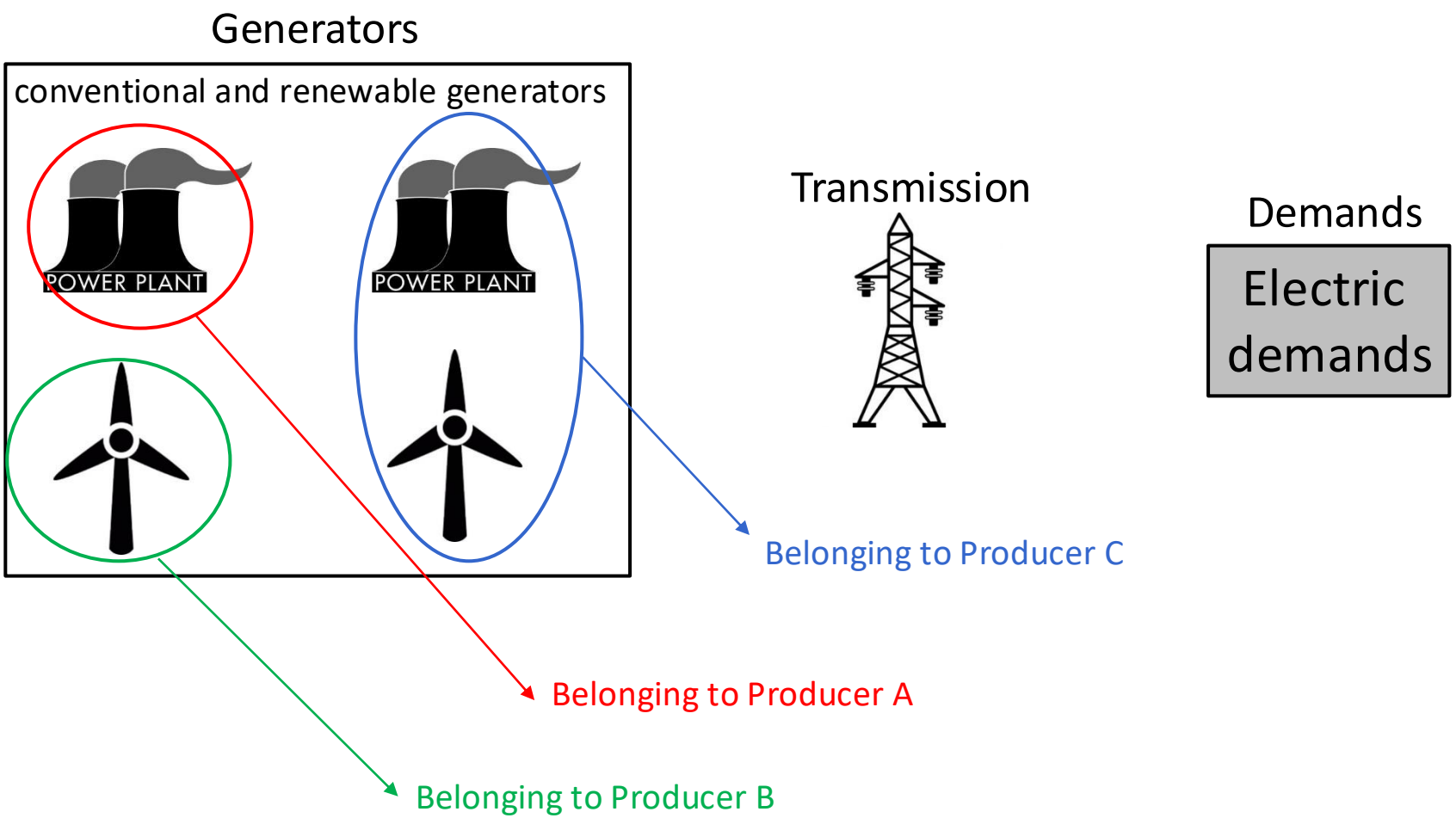
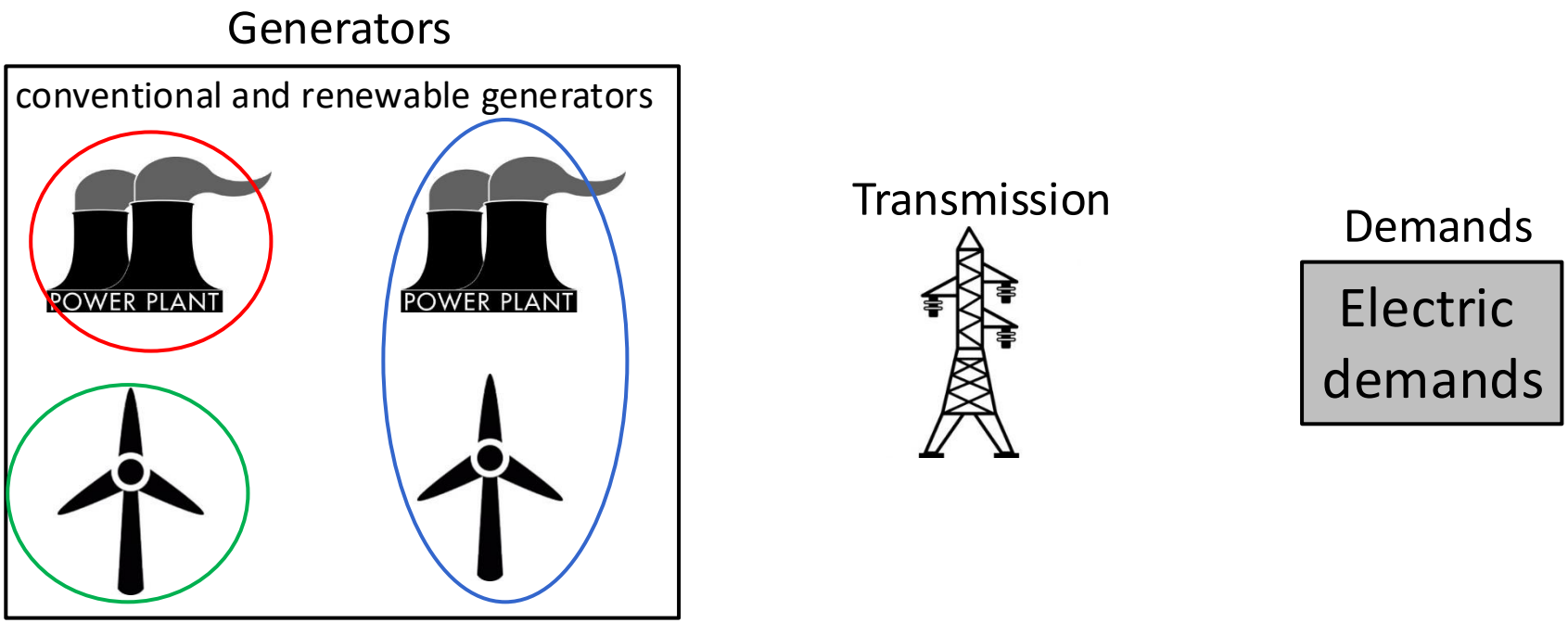
- **Electricity Markets**

Electricity Markets



Electricity Markets



Electricity Market:

- Each producer aims to maximize their **<u>own profit</u>** by making optimal operational and planning decisions.
- Unlike centralized systems, there are **<u>multiple</u>** decision-makers involved.

Electricity Markets

How can an electricity market be operated where each market participant (e.g., generators) makes their own profit-maximizing decisions?

Electricity Markets

How can an electricity market be operated where each market participant (e.g., generators) makes their own profit-maximizing decisions?

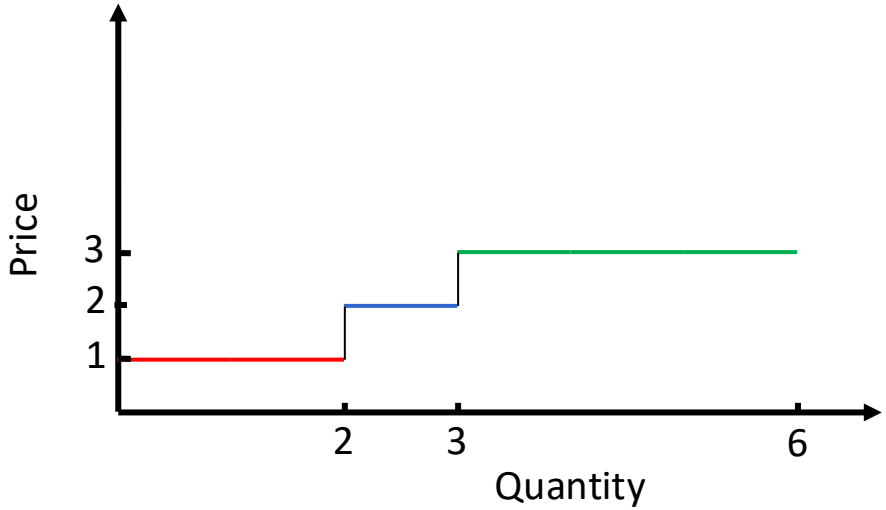
For clarity, let us consider an apple market!

Market

Assume there is an apple market with multiple sellers and buyers:

Sellers:

- Seller A sells 2 apples at a price of $1 each.
- Seller B sells 1 apple at a price of $2.
- Seller C sells 3 apples at a price of $3 each.



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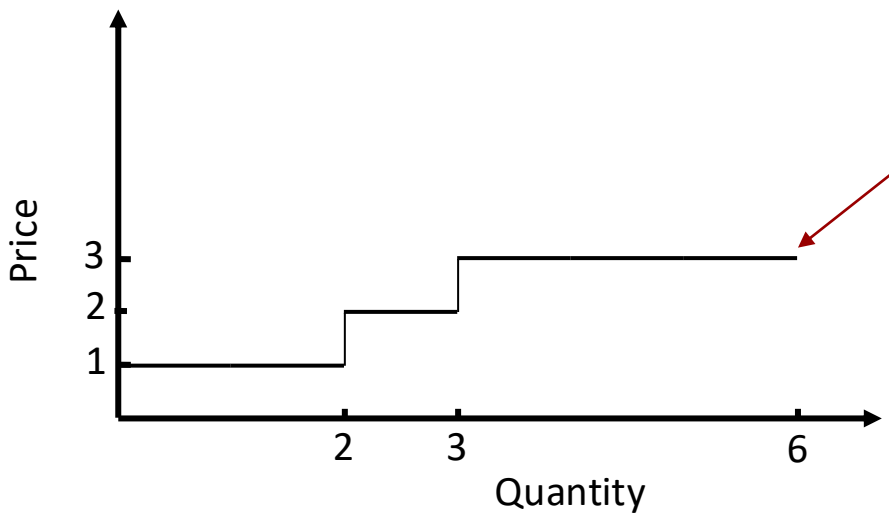


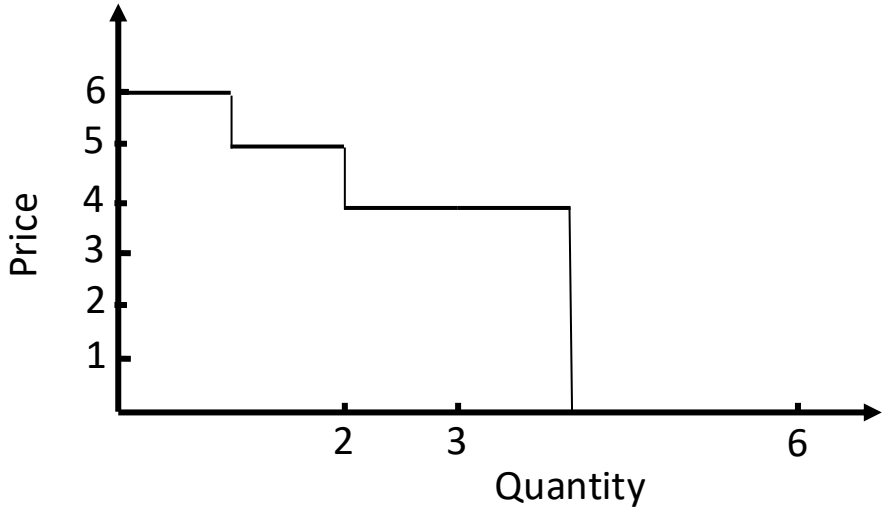
Supply curve:
a non-decreasing curve where sellers are ranked based on the least-cost "merit order principle"!

Market

Assume there is an apple market with multiple sellers and buyers:

Buyers:

- Buyer A buys 1 apple at a price of $6.
- Buyer B buys 1 apple at a price of $5.
- Buyer C buys 2 apples at a price of $4 each.



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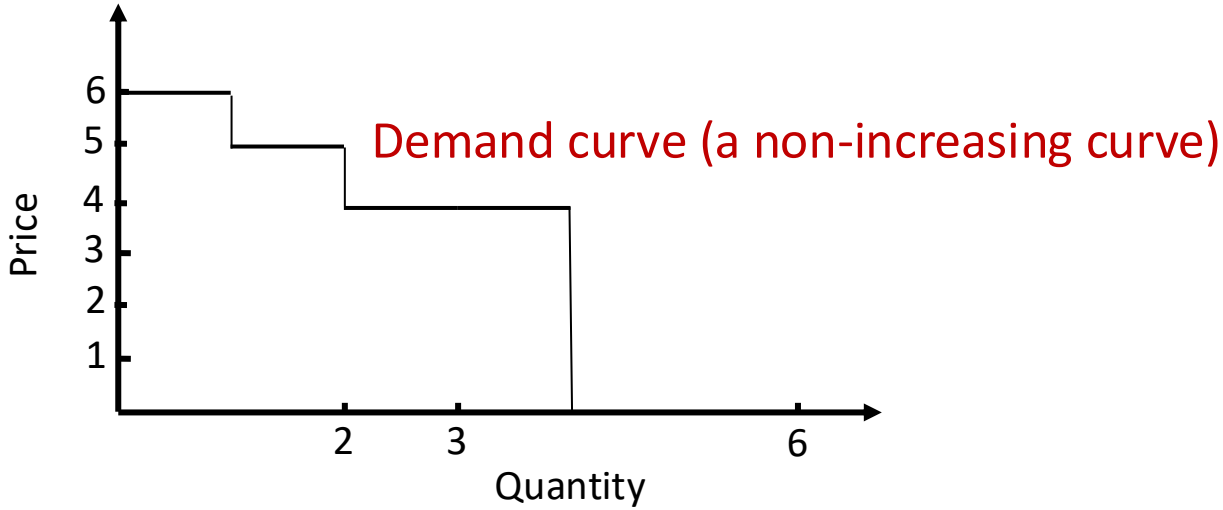

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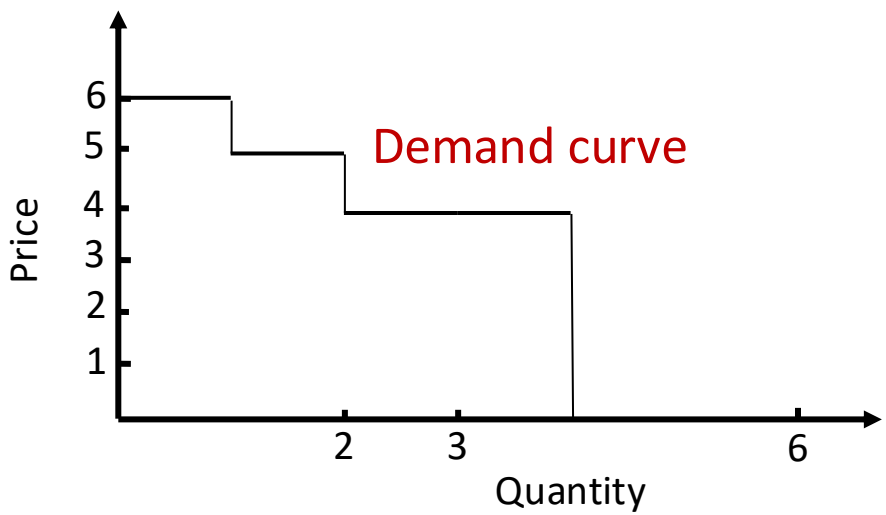
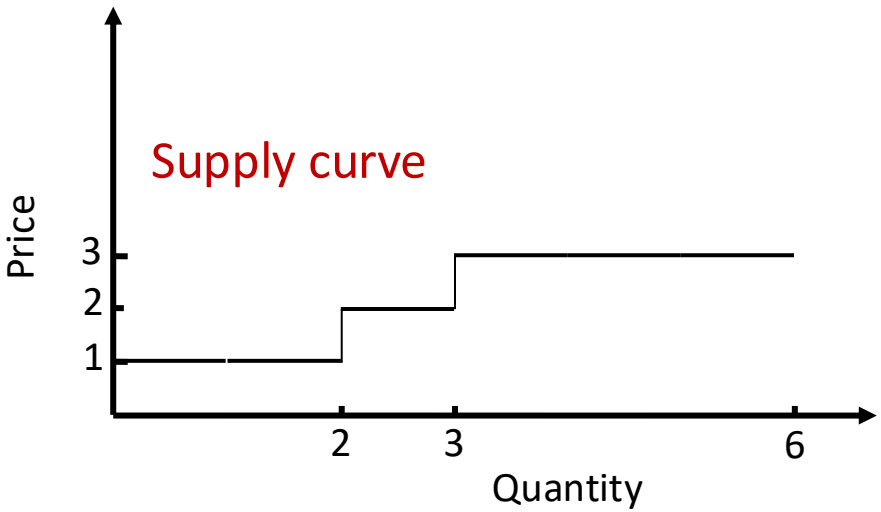
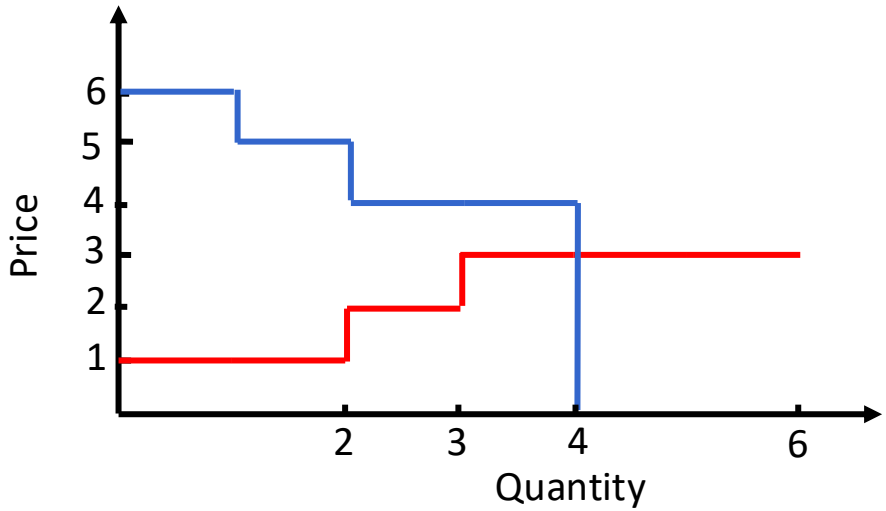




Now that we have both supply and demand curves, let's determine their <u>intersection</u>!

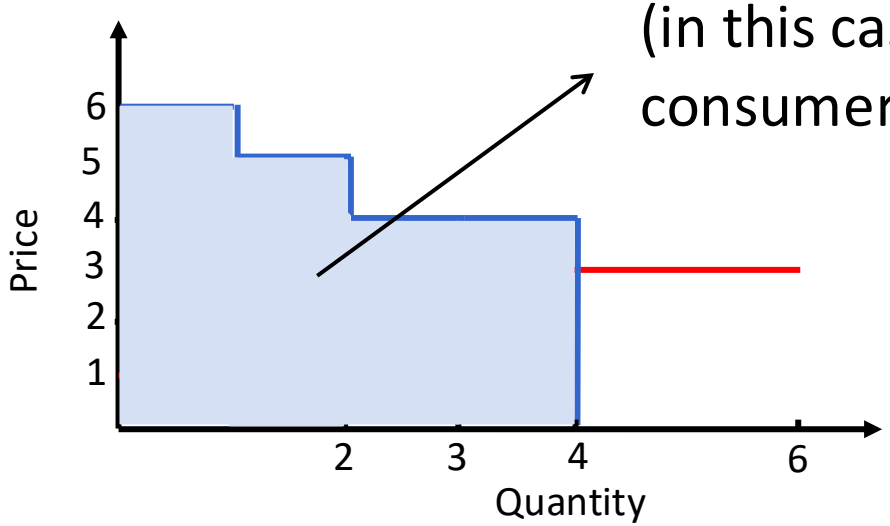
Market

Supply-demand curve



This is known as the "**market-clearing**" procedure, which is carried out by a non-profit entity, the <u>market operator</u>!

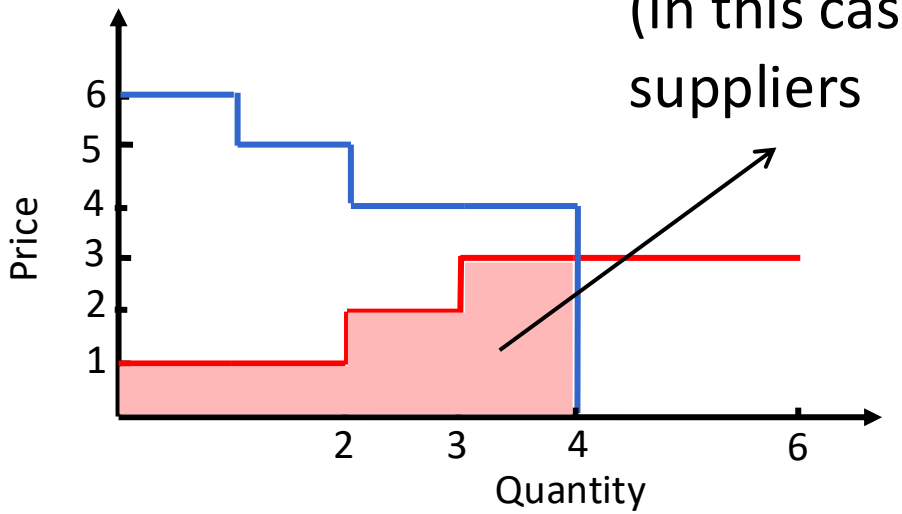
Market



The total value of commodities (in this case, apples) for the consumers

Market

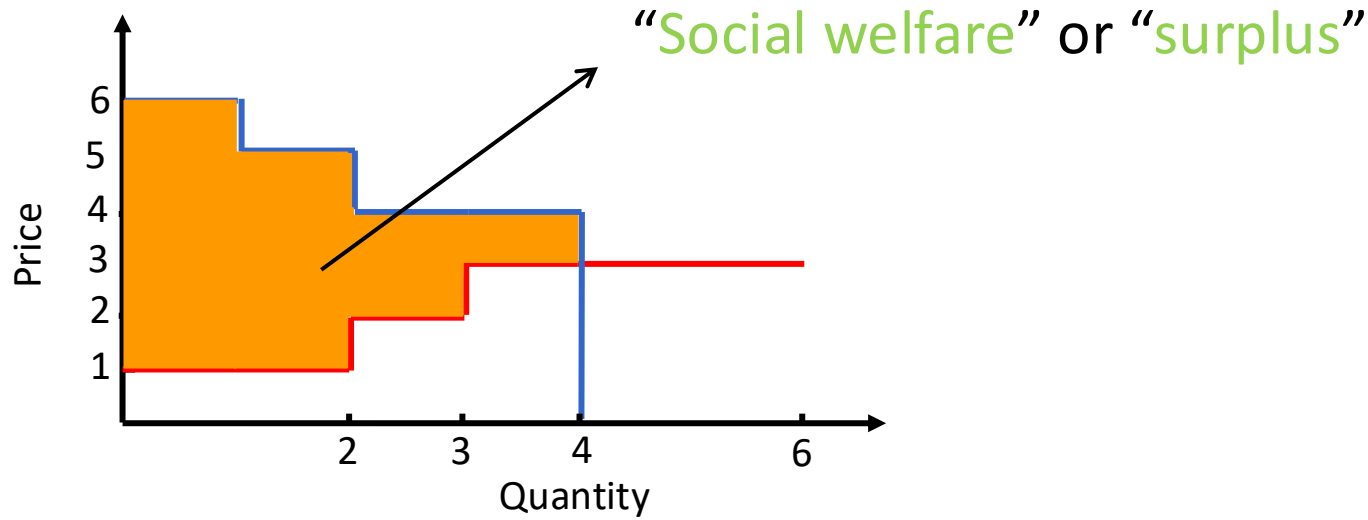
The total cost of commodities (in this case, apples) for the suppliers



Market

The market operator clears the market by ***maximizing*** "social welfare", which is the area between the supply and demand curves!

Market

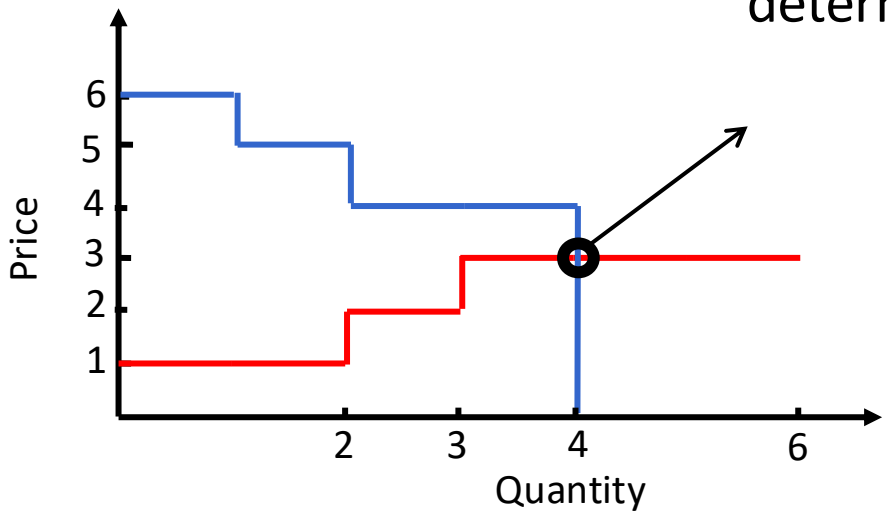


The market operator clears the market by ***maximizing*** "social welfare", which is the area between the supply and demand curves!

- By maximizing social welfare, the total value of commodities for consumers is maximized, while the total cost for suppliers is minimized—resulting in satisfaction for both the demand and supply sides!

Market

Equilibrium point: The point where the market-clearing **price** and **quantity** are determined.



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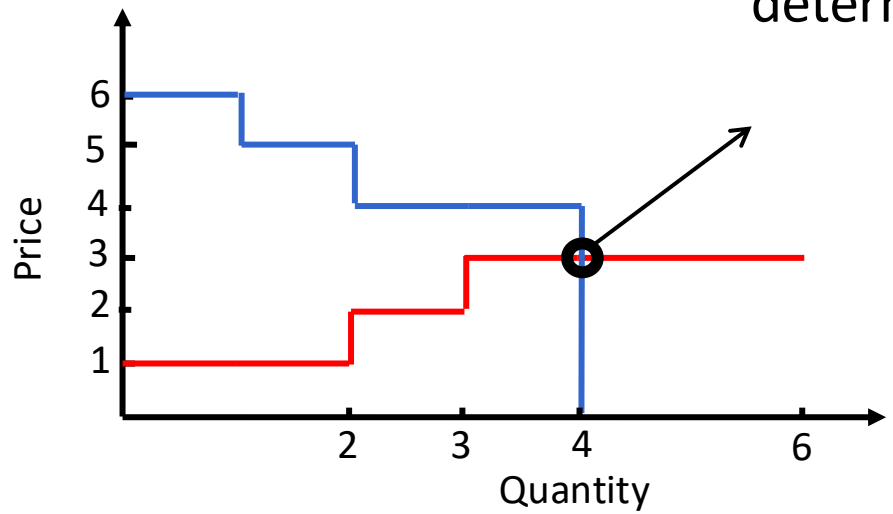

- At the equilibrium point, the social welfare is maximized.

Market

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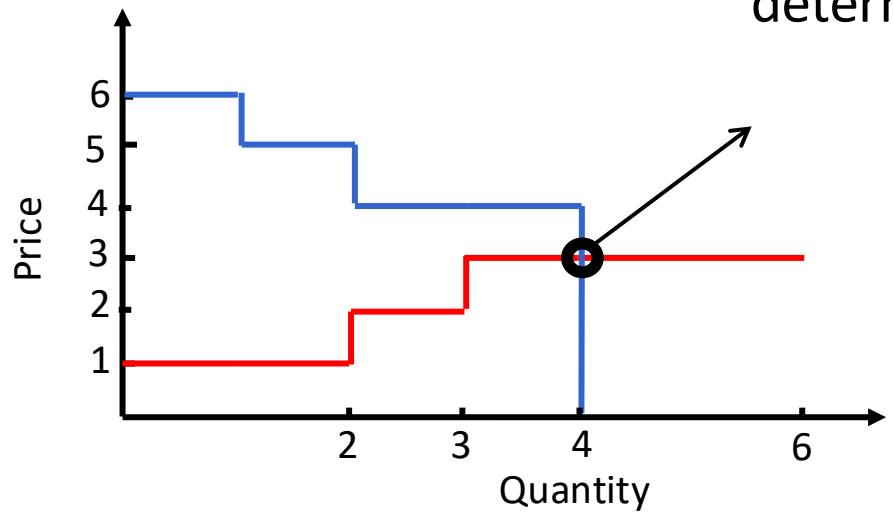
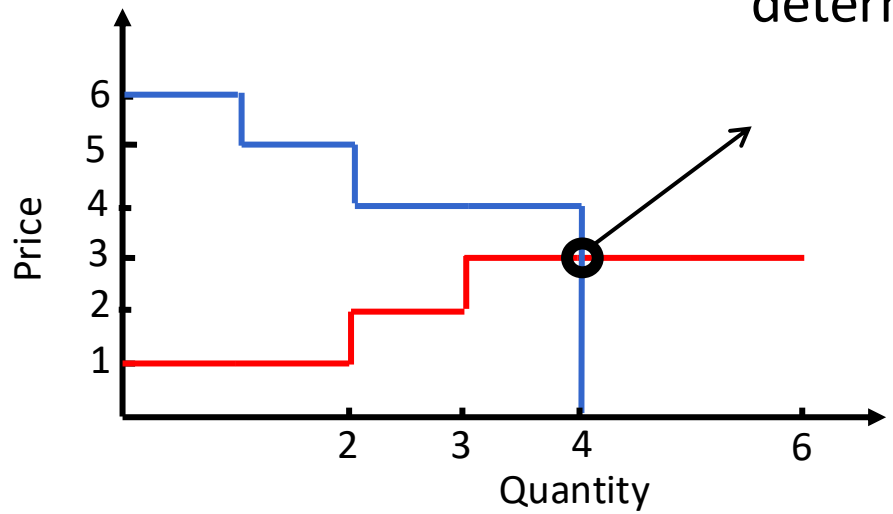


- At the equilibrium point, the social welfare is maximized.
- Based on the equilibrium point, four apples are traded at a price of $3 each (uniform pricing)!

Market

Question: Based on uniform pricing, does any buyer (or seller) necessarily pay (or receive) the price they submitted to the market?

Equilibrium point: The point where the market-clearing **price** and **quantity** are determined.



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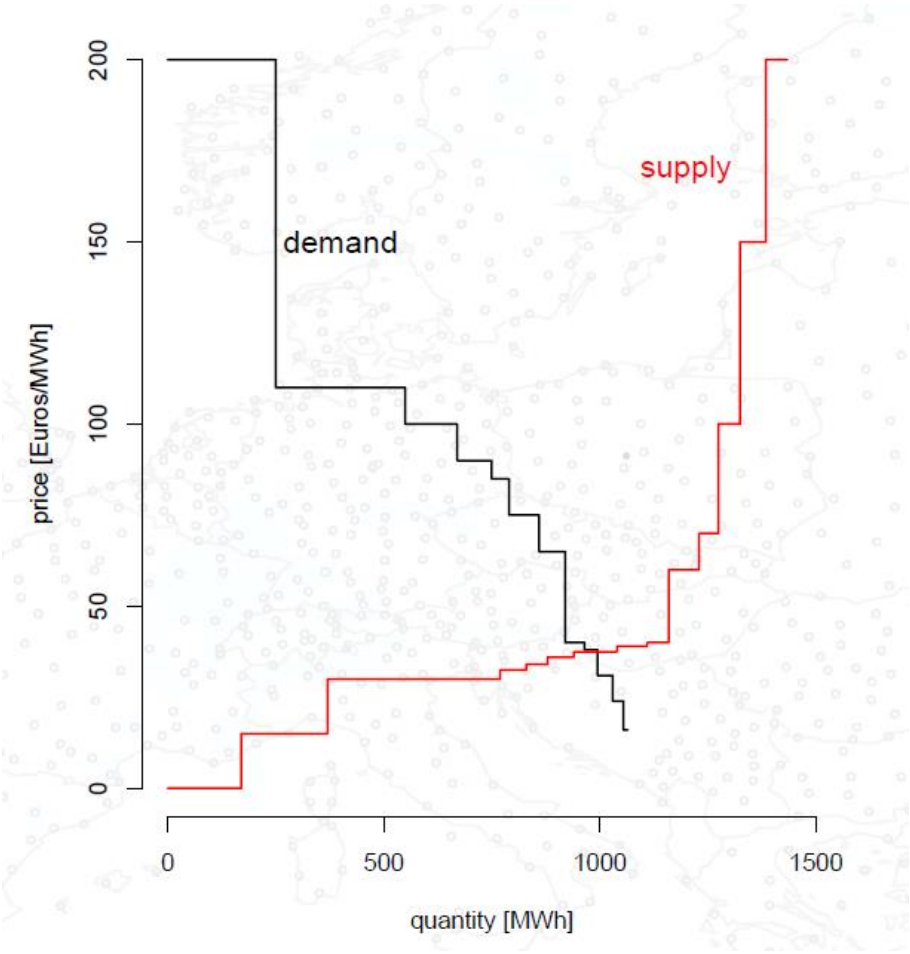
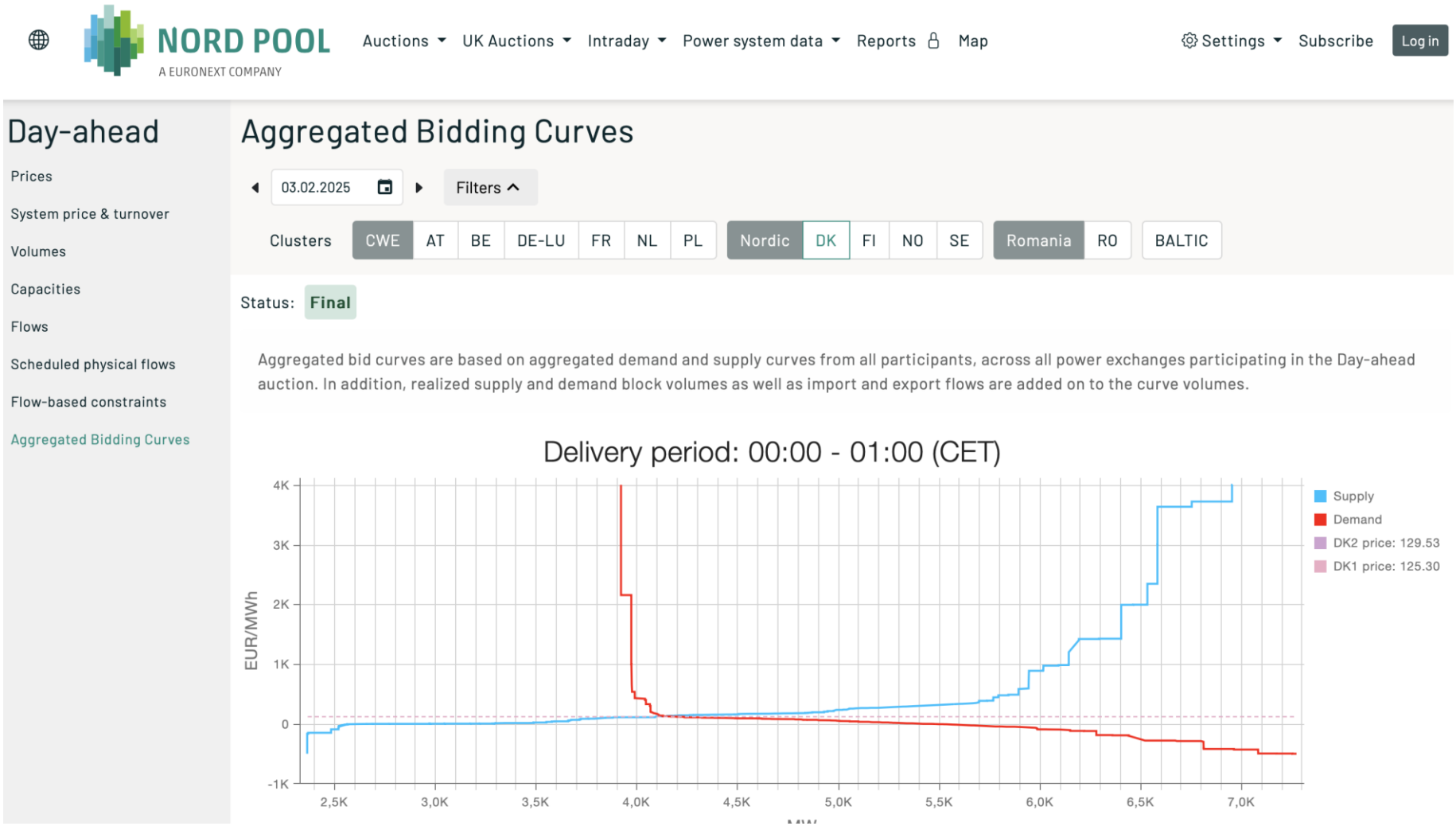
Electricity Markets

The electricity market also trades a commodity (like the apple market), and its clearing algorithm is similar!



Let’s check supply-demand curves in Nord Pool



Link: https://data.nordpoolgroup.com/auction/day-ahead/aggregated-bid-curves?deliveryDate=2025-02-03&bidCurveArea=DK

Electricity Markets

Discussion:

What aspects of an electricity market differentiate it from markets for other commodities, such as the apple market?

Electricity Markets

Discussion:

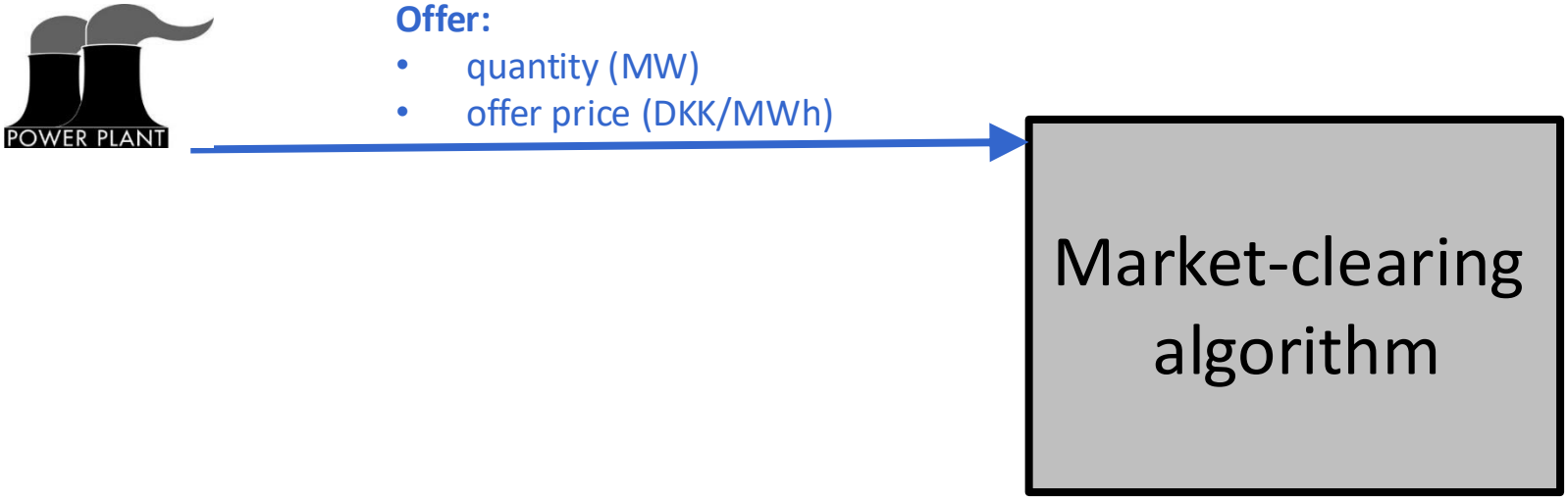
What aspects of an electricity market differentiate it from markets for other commodities, such as the apple market?

- The electricity market-clearing algorithm must take into account the physical laws of electric networks (e.g., Kirchhoff's circuit laws),
- Electricity is a non-storable commodity on a large scale,
- Electricity demand is typically highly inelastic to price, though this is changing,
- …

Electricity Markets

Market-clearing algorithm

Electricity Markets



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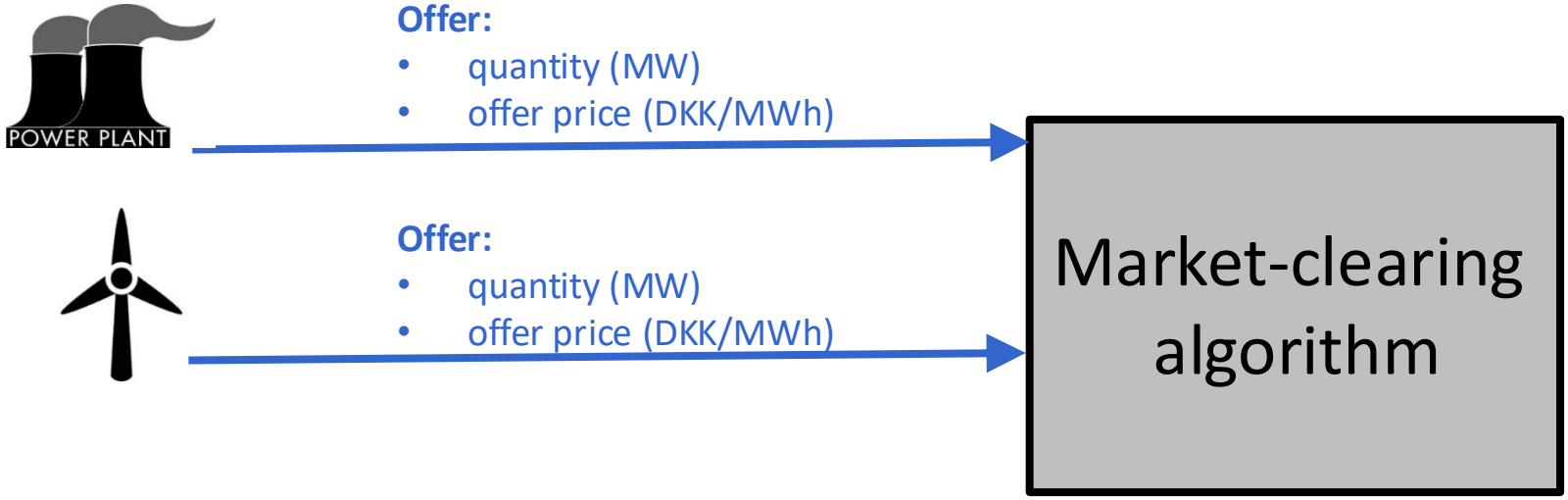

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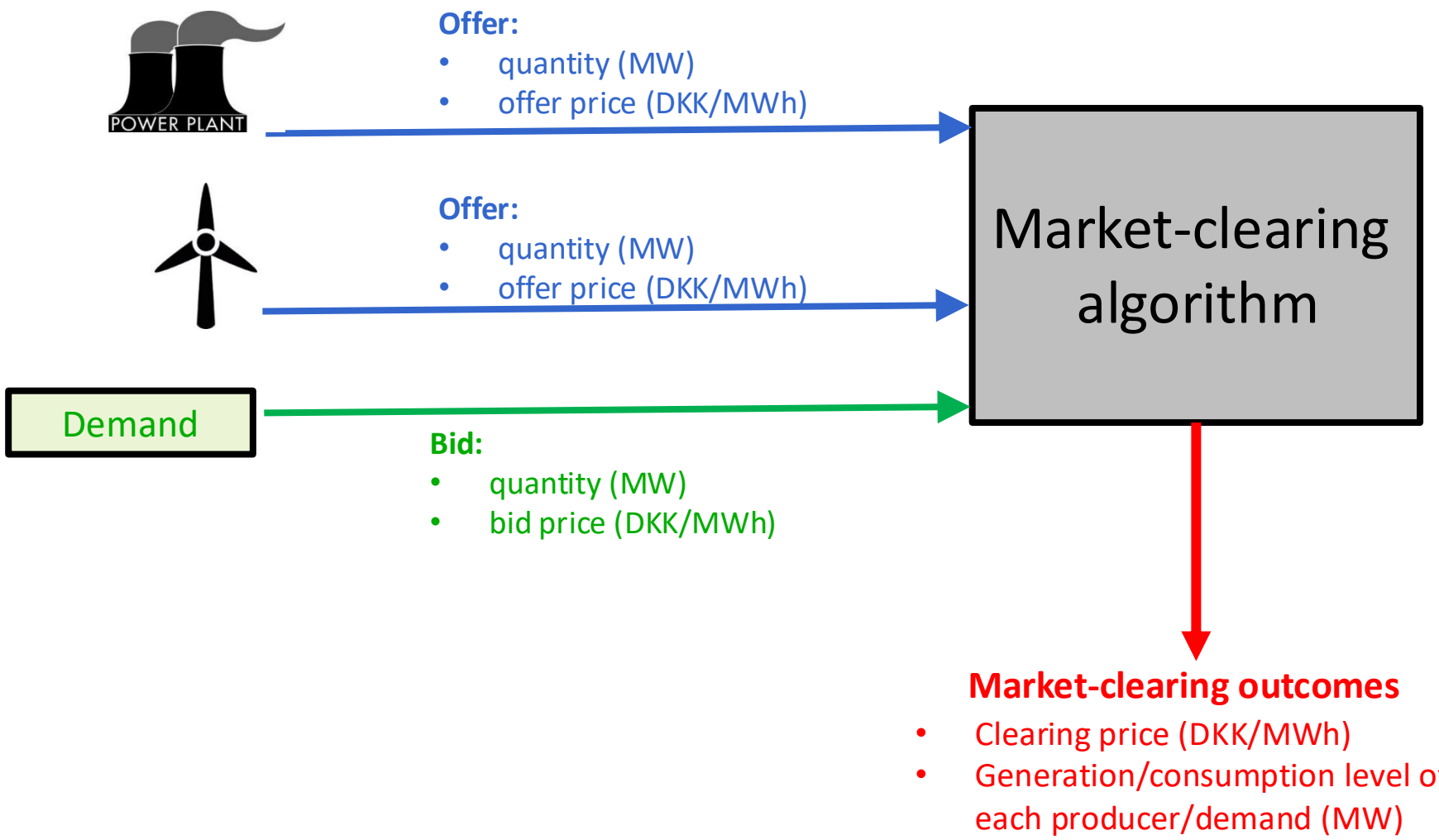

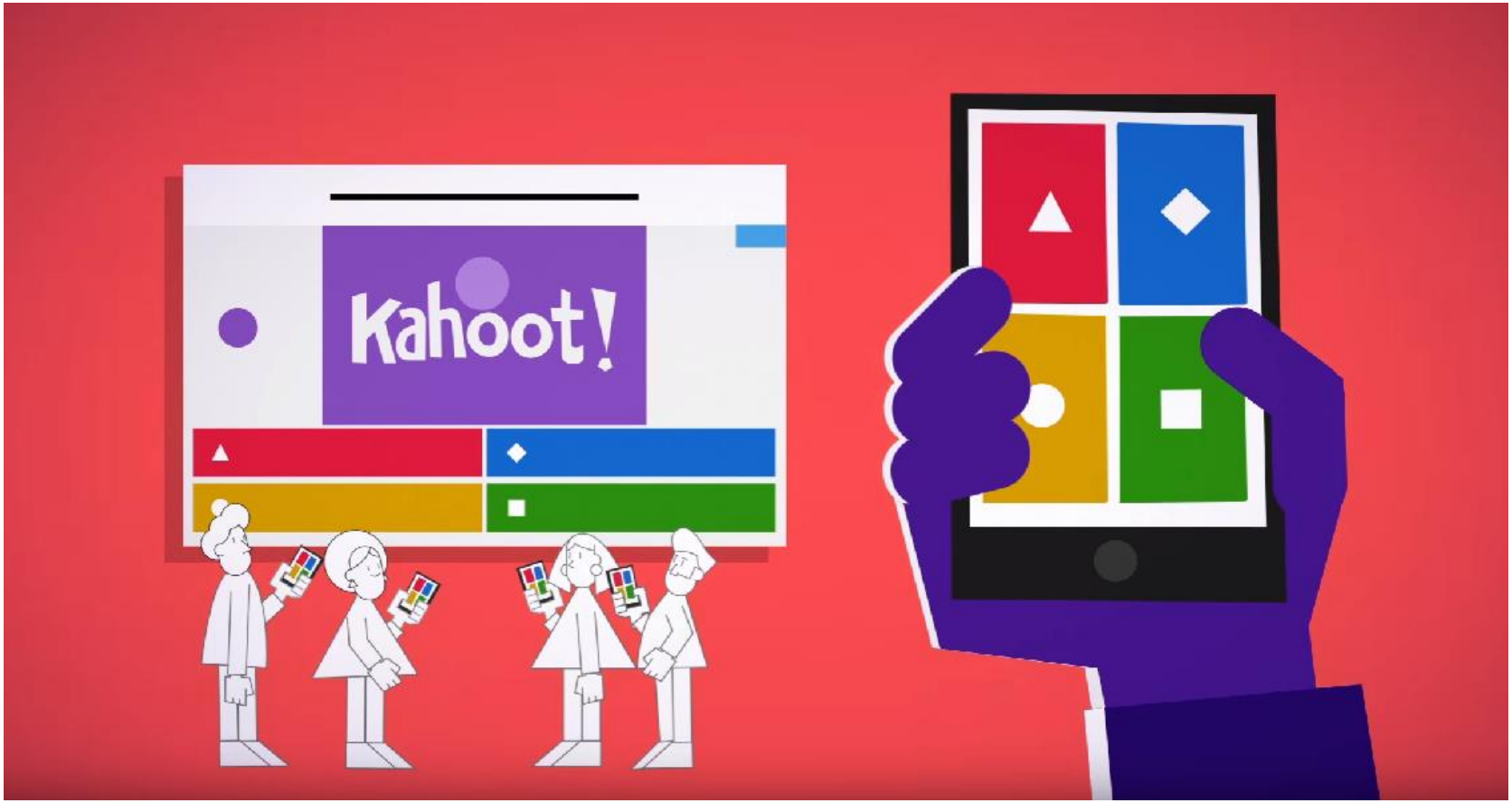
Kahoot!

Thanks for your attention!

Email: jalal@dtu.dk